南方科技大学
SOUTHERN UNIVERSITY OF SCIENCE AND TECHNOLOGY
电子与电气工程系
DEPARTMENT OF ELECTRONIC AND ELECTRICAL ENGINEERING

Fundamentals of Optics

Lecturer

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Q&A time: 2-4 pm on Monday

Room 239, South Tower, College of Engineering

Rules

- **Final exam: 30%**
- **Mid-term exam: 30%**
- **Homework: 25%**
- **Attendance: 15%**

Reference books

Optics, by Eugene Hecht
光学，赵凯华、钟锡华

CHAPTER 3
THE PROPAGATION OF LIGHT

3.1 Introduction

Reflection (反射)

Transmission (透射)

Refraction (折射)

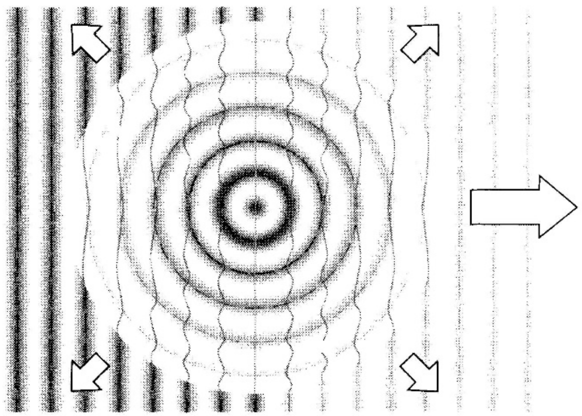
Scattering

- **Scattering:** the absorption and prompt re-emission of EM-radiation by electrons associated with atoms and molecules.

  The process of transmission, reflection, and refraction are macroscopic manifestations of scattering occurring on a submicroscopic level.

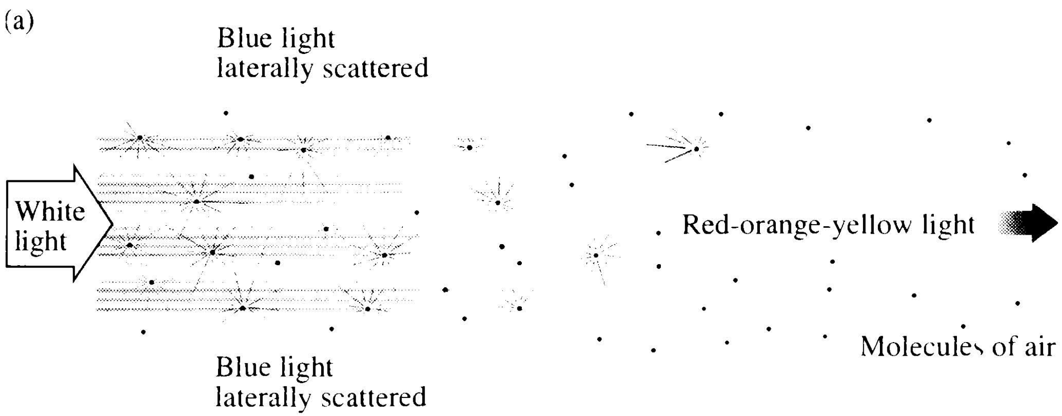
3.2 Rayleigh Scattering

- Consider a wisp of air (particles smaller than a wavelength)
  - ✓ In the visible, these molecules have no resonances. It means none of them can be raised into an excited state by absorbing a quantum of light, and the gas is therefore transparent.
  - ✓ Instead, each molecule behaves as a little oscillator whose electron cloud can be driven into a ground-state vibration by an incoming photon.
  - ✓ Immediately upon being set vibrating, the molecule initiates the re-emission of light.
  - ✓ A photon is absorbed, and without delay another photon of the same frequency is emitted.
- The light is **elastically scattered**



Wave dependence of scattering

All the molecules have electronic resonances in the UV.
The closer the driving frequency is to a resonance, the more vigorously the oscillator responds



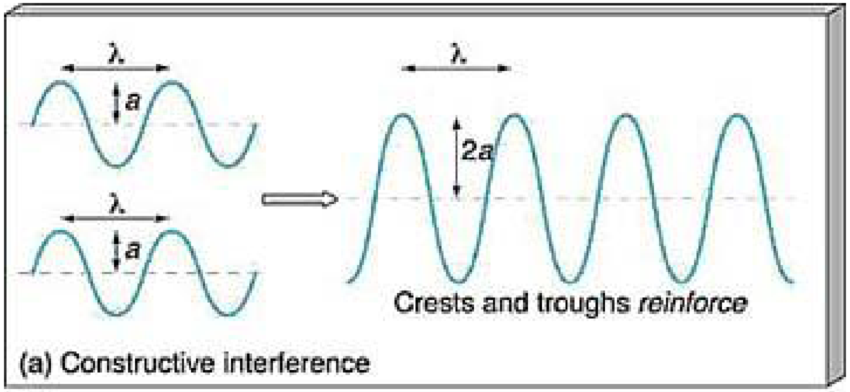
Intensity of the light scattered: $I \propto \dfrac{1}{\lambda^4}$

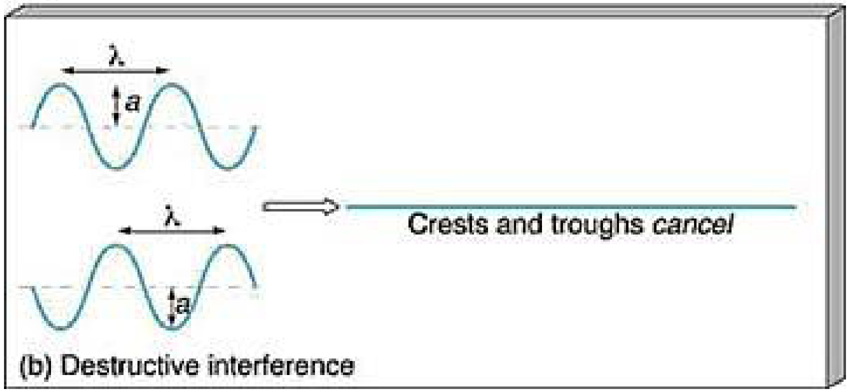
Case: $I_{\textbf{(blue light 430nm)}} \approx 6 \times I_{\textbf{(red light 680nm)}}$

3.2.1 Scattering and interference

- Interference: the superposition of two or more waves producing a resultant disturbance that is the sum of the overlapping wave contributions.

- Total constructive interference: in-phase

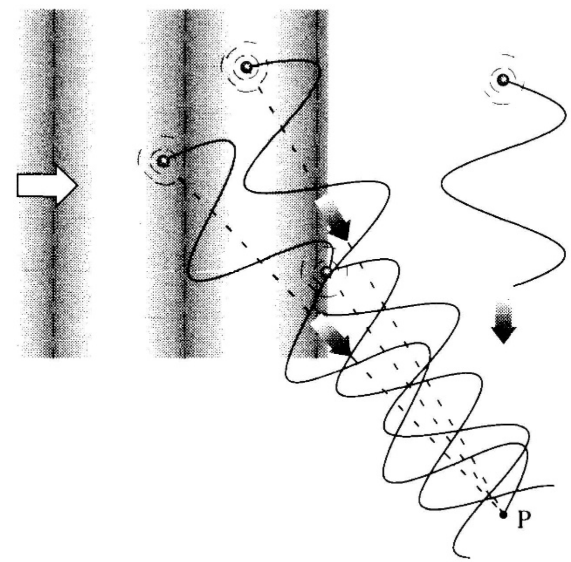




- Total destructive interference: 180° out-of-phase

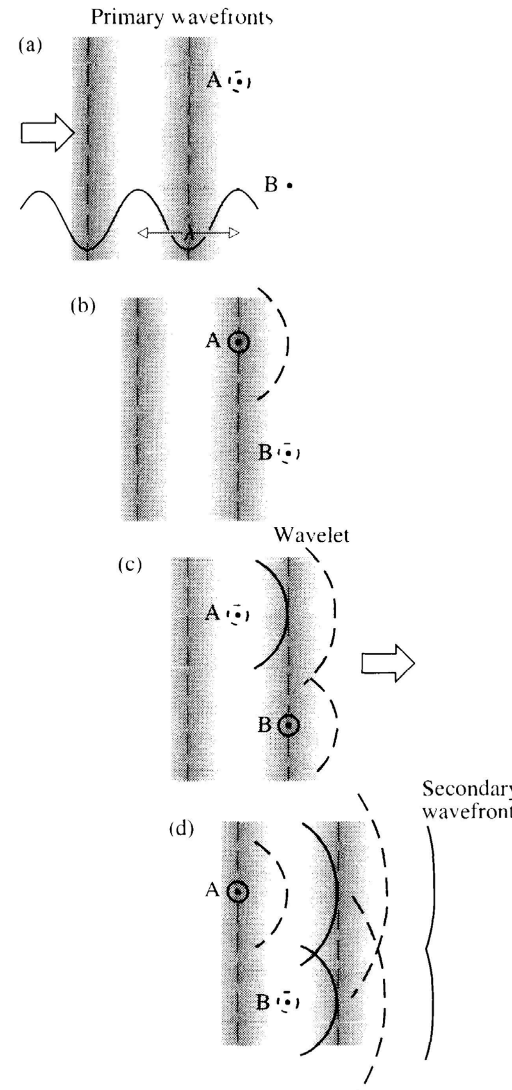
Tenuous gas: molecules randomly arrayed in space

The wavelets arriving at a lateral point P have a jumble of different phases and tend not to interfere in a sustained constructive fashion.



Random, widely spaced scatters driven by an incident primary wave emit wavelets that are essentially independent of one another in all direction except forward. **Laterally scattered light, unimpeded by interference, streams out of the beam**.

All the scattered wavelets add **constructively** with each other in the **forward direction**.

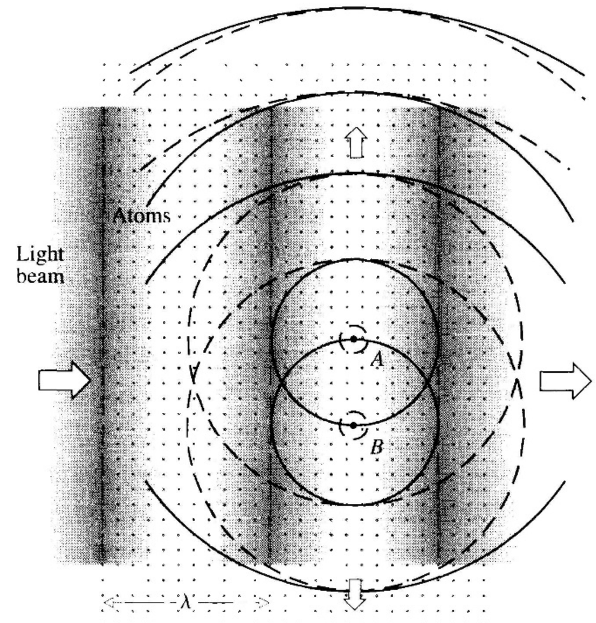


It is also true for dense media

3.2.2 The transmission of light through dense media

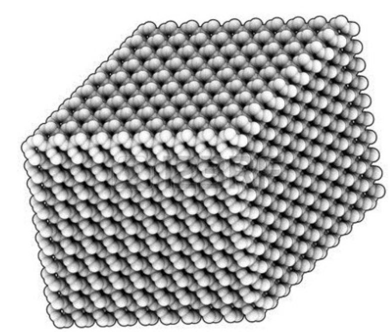
Little or no light ends up scattered laterally or backwards in a dense homogeneous medium.

Interference produces a redistribution of energy, out of the regions where it's destructive into the regions where it's constructive.



A and B are separated by $\lambda/2$

Example: Good crystals (quartz) scatter even more faintly



3.2.3 Transmission and the index of refraction

- Secondary wave: the scattered wavelets combined in-phase in the forward direction.
- Primary wave: the incident electromagnetic wave.
- Transmitted wave: primary wave + secondary wave

The primary and secondary electromagnetic waves propagate through the interatomic void with the speed c.

But the velocity in medium $v = \dfrac{c}{n}$

Source: the phase relationship between the secondary and primary waves

Phase shift

Classical forced oscillator:

Opposing restoring force

$$\boxed{q_e E(t) - m_e {\omega_0}^2 x - m_e \gamma \frac{dx}{dt} = m_e \frac{d^2 x}{dt^2}}$$

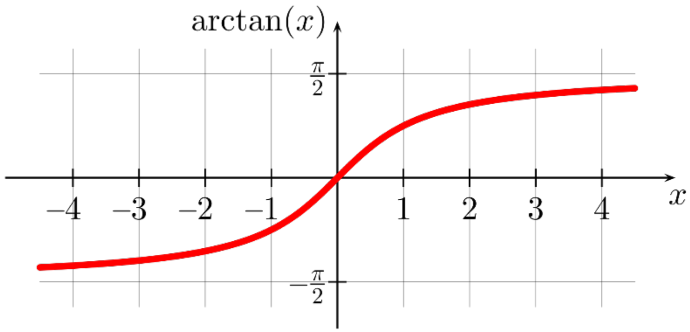
Driving force

Damping force, energy lost when the forced oscillators re-radiate.

Let $E(t) = E_0 \boldsymbol{e^{i\omega t}}$

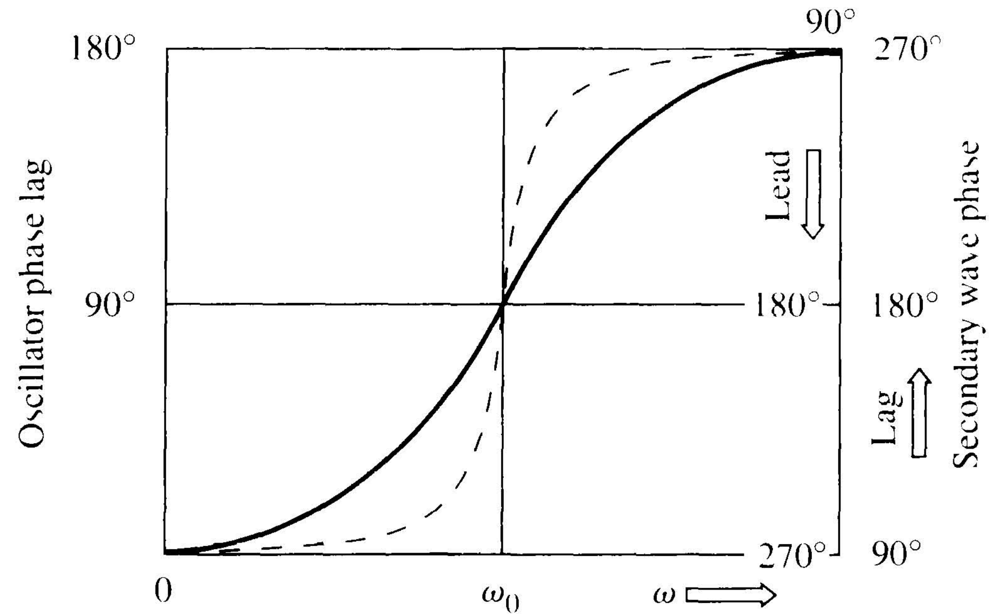
The solution: $x = x_0 \boldsymbol{e^{i(\omega t - \alpha)}}$

$$x_0 = \frac{q_e E_0}{m_e} \frac{1}{[({\omega_0}^2 - \omega^2)^2 + \gamma^2 \omega^2]^{1/2}}$$



$$\alpha = \tan^{-1}\left[\frac{\gamma\omega}{{\omega_0}^2 - \omega^2}\right]$$

$$\tan\alpha = \frac{\gamma\omega}{{\omega_0}^2 - \omega^2}$$



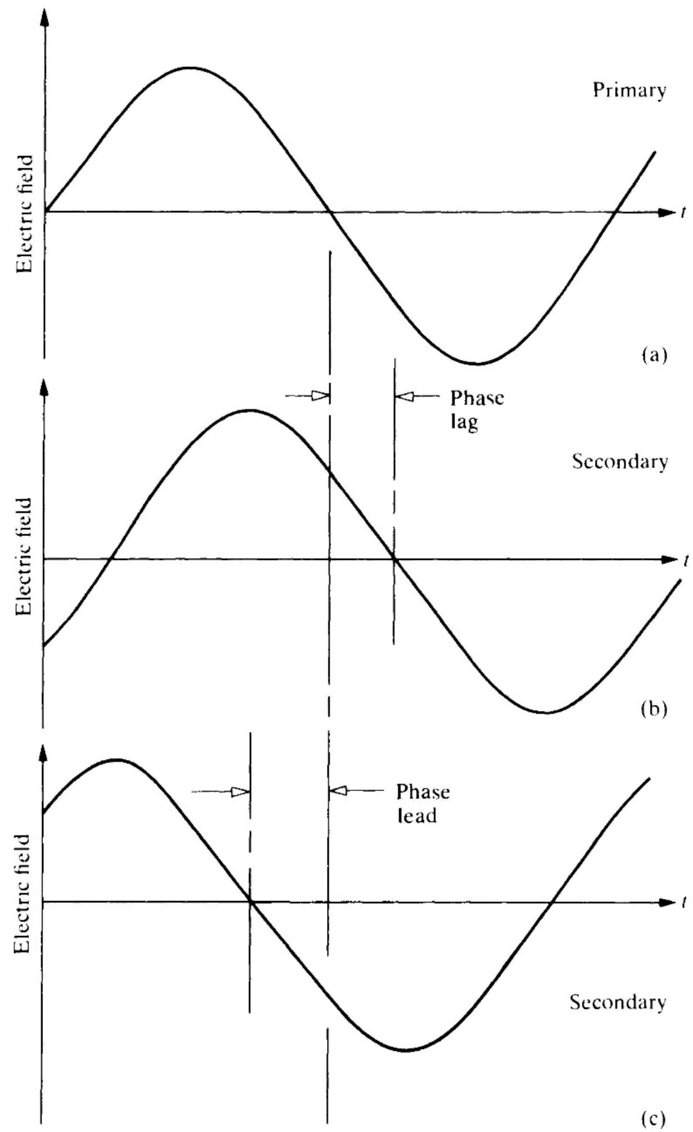
The resultant secondary wave itself lags the oscillators by 90°. (Homework: Problem 4.5)

- ❑At frequency below resonance: the secondary wave lags the primary by some amount between approximately 90° and 180°
- ❑At frequency above resonance: the secondary wave lags the primary by some amount between approximately 180° and 270°
- ❑Note: a phase lag of δ≥180° is equivalent to a phase lead of 360°−δ

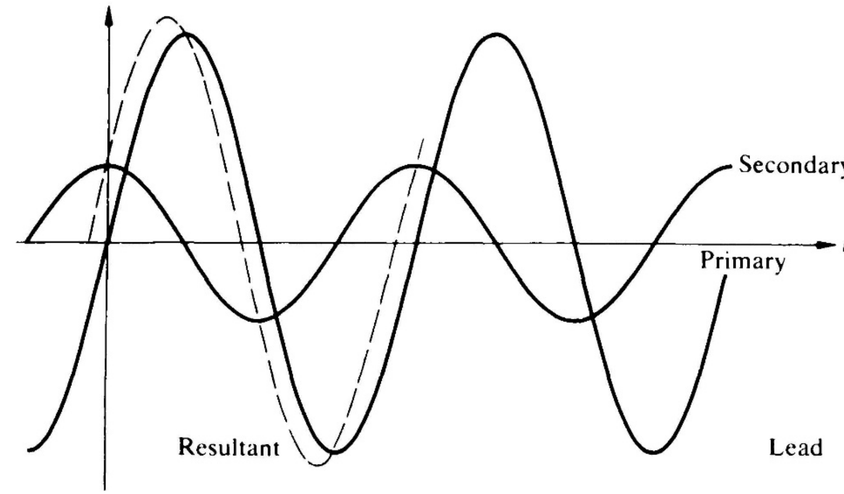
- A primary wave

- The secondary lags the primary- it takes longer to reach any given value.

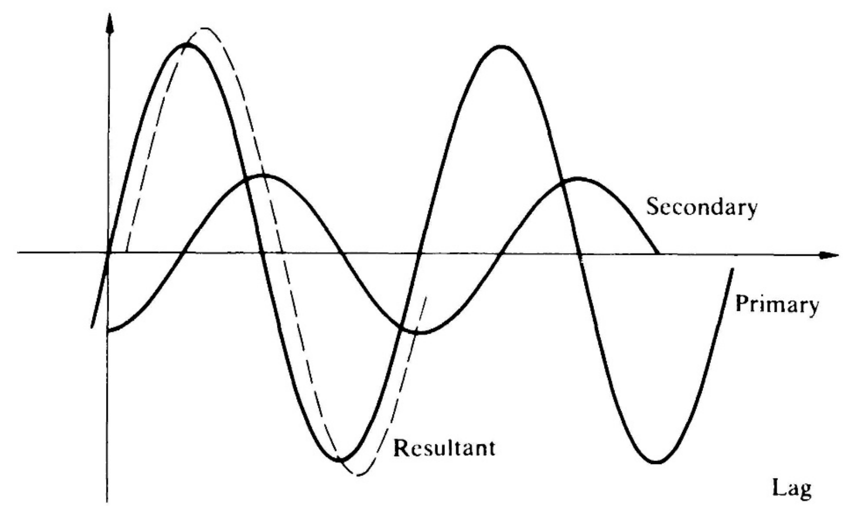
- The secondary wave reaches any given value before the primary; that is, it leads.



- When the secondary wave leads the primary, the resultant transmitted wave leads it by some amount.
  - ✓ $\omega>\omega_0$: the transmitted wave leads the free-space wave
  - ✓ A phase lead yields an increase in speed, $v>c$ and $n<1$



- When the secondary wave lags the primary, the resultant transmitted wave lags it by some amount.
  - ✓ $\omega<\omega_0$: the transmitted wave lags the free-space wave
  - ✓ A phase lag corresponds to a reduction in speed, $v<c$ and $n>1$



Homework

Problem 4.5

4.5 Imagine that we have a nonabsorbing glass plate of index n and thickness Δy, which stands between a source S and an observer P.

(a) If the unobstructed wave (without the plate present) is $E_u = E_0 \exp i\omega(t - y/c)$, show that with the plate in place the observer sees a wave

$$E_p = E_0 \exp i\omega[t - (n - 1)\,\Delta y/c - y/c]$$

(b) Show that if either $n \approx 1$ or Δy is very small, then

$$E_p = E_u + \frac{\omega(n-1)\,\Delta y}{c} E_u e^{-i\pi/2}$$

The second term on the right may be envisioned as the field arising from the oscillators in the plate.